

**REPORT
OF
ARTIFICIAL INTELLIGENCE & MACHINE LEARNING**

Submitted To

**School of Computer Science and Engineering
IILM University, Greater Noida, U.P.**

**In partial fulfilment of the requirement of degree of
B.Tech CSE**



By

Name of Student: ROUSHAN KANT SINGH

Roll Number of Student: 25SCS1003002793

Batch: 2025-29

CANDIDATE’S DECLARATION

I, **ROUSHAN KANT SINGH**, do hereby declare that the internship report titled **“Artificial Intelligence & Machine Learning”** has been completed by me to fulfil the requirement for the award of the degree of **Bachelor of Technology in Computer Science and Engineering**. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature: _____

Student Name: ROUSHAN KANT SINGH

Roll No.: 25SCS1003002793

Date: 30th August 2026

Place: Greater Noida, U.P.

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to extend my sincere gratitude to my internship mentor, **Dr. Anurag Shrivastava** (Program Manager, Codec Technologies Pvt. Ltd.), for his invaluable guidance, continuous support, and technical feedback throughout the internship tenure.

I also express my deep appreciation to the faculty members and Dean of the **School of Computer Science and Engineering, IILM University, Greater Noida**, for providing this platform and fostering an inspiring academic environment.

I express my gratitude to my parents for their blessings.

Signature: _____

Student Name: ROUSHAN KANT SINGH

Roll No.: 25SCS1003002793

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CERTIFICATE OF INTERNSHIP

This certificate is awarded to
ROUSHAN KANT SINGH

From Codec Technologies Pvt. Ltd.

In recognition of his/her efforts and achievements in completing the 1 Month AICTE & ICAC

Approved internship program as

Artificial Intelligence Intern

Conducted from 15/06/2026 to 15/07/2026



Google for Education
Partner



AICTE ID - CORPORATE6759d549ce59e1733940553

A handwritten signature in black ink, appearing to read 'Anurag'.

Dr. Anurag Shrivastava
Program Manager

INTERNSHIP OFFER LETTER

Dear ROUSHAN KANTSINGH,

We are pleased to offer you a 1 Month Internship as Project Intern at Codec Technologies, a global platform dedicated to empowering learners and connecting diverse talent to create meaningful career opportunities worldwide. Codec Technologies delivers dedicated IT and business consultancy services across 27+ countries, empowering global innovation and strategic growth.

Internship Details:

- Designation : Artificial Intelligence Intern
- Location & Mode: Pan India, Hybrid
- Duration : 15/06/2026 to 15/07/2026
- Reporting to : Assigned Project Head(s)

This program provides an immersive experience, enabling you to gain industry-level knowledge and acquire valuable skills. Exceptional interns may also qualify for a scholarship award based on weekly performance reviews conducted by project supervisor.

If you accept this offer and the terms above, please complete your assigned projects and training tasks on our platform and via email.

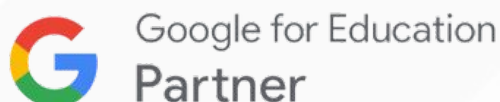
Best regards,

Sincerely,

Dr. Anurag Shrivastava,
Codec Technologies India
NCS ID: E19E86-0116588288923
Date Of Issue - 15/06/2026



Dr. Anurag Shrivastava
Talent Acquisition Manager



CHAPTER 4: PROJECT DESCRIPTION

4.1 Introduction

The one-month Artificial Intelligence Internship Program conducted by **Codec Technologies Pvt. Ltd.** in association with the **National Internship Portal (Ministry of Education)** and **Google for Education** was structured to provide comprehensive practical immersion in Artificial Intelligence (AI), Machine Learning (ML), Deep Learning (DL), Natural Language Processing (NLP), and Reinforcement Learning (RL).

Throughout the tenure of this program (15/06/2026 to 15/07/2026), practical software engineering competencies were established across the end-to-end data lifecycle—spanning exploratory data analysis, time-series forecasting, convolutional neural network (CNN) architectures for computer vision, natural language syntactic parsing, sequential decision-making policy optimization via Q-learning, and applied capstone engineering implementations.

4.2 Organization Profile

Codec Technologies Pvt. Ltd. is a specialized technology software and industrial training enterprise approved by AICTE and ICAC. Codec Technologies focuses on delivering professional upskilling and applied industrial training across emerging computing paradigms. Partnered with Google for Education and operating through the National Internship Portal, the organization provides structured curricula, industry-grade project mentorship, and standardized evaluation frameworks.

- **AICTE Verification ID:** CORPORATE6759d549ce59e1733940553
- **NCS Identification:** E19E86-0116588288923
- **Quality Certification:** ISO 9001:2015 Certified
- **Program Leadership:** Dr. Anurag Shrivastava (Program Manager / Talent Acquisition Manager)
- **Institutional Partnerships:** Google for Education Partner, AICTE & ICAC Approved

4.3 Problem Statement

Modern digital applications generate heterogeneous, multi-modal data streams comprising continuous financial time-series, spatio-temporal meteorological sensor records, high-dimensional visual matrices, and unstructured text. Conventional rule-based software systems fail to extract non-linear temporal correlations or generalize across dynamic, noisy environments.

The objective of this project is to develop and optimize modular machine learning and deep learning architectures capable of:

1. Ingesting, cleaning, and preprocessing multi-variate continuous time-series and environmental sensor data.
2. Modeling sequential temporal dependencies in equity markets to predict future stock valuation trends.
3. Extracting meteorological feature correlations to classify and forecast atmospheric weather conditions and precipitation events.
4. Quantifying system generalization performance through rigorous evaluation metrics (RMSE, MAE, R^2 , Accuracy, Precision, Recall, F1-Score).

4.4 Project Objectives

- To establish advanced programming workflows in Python utilizing NumPy, Pandas, Matplotlib, Seaborn, and Scikit-Learn.
- To design and deploy a Deep Recurrent Neural Network utilizing Long Short-Term Memory (LSTM) cells for financial equity price forecasting.
- To construct an end-to-end meteorological data analytics pipeline and benchmark ensemble classification algorithms for rainfall prediction.
- To master feature engineering techniques including sliding window framing, min-max scaling, categorical imputation, and SMOTE class balancing.
- To integrate all theoretical principles and practical modules into comprehensive, verifiable Capstone Projects.

4.5 Scope of the Project

The project scope covers key areas in applied artificial intelligence:

- **Time-Series Forecasting & Deep Sequence Modeling:** Recurrent state updates, vanishing gradient mitigation, and multi-step lookback windows.
- **Meteorological Data Analytics & Predictive Classification:** Atmospheric parameter exploration, multi-collinearity reduction, and ensemble decision boundaries.
- **Machine Learning Lifecycle Management:** Data cleaning, transformation, model evaluation, and hyperparameter tuning.

4.6 Technologies and Tools Used

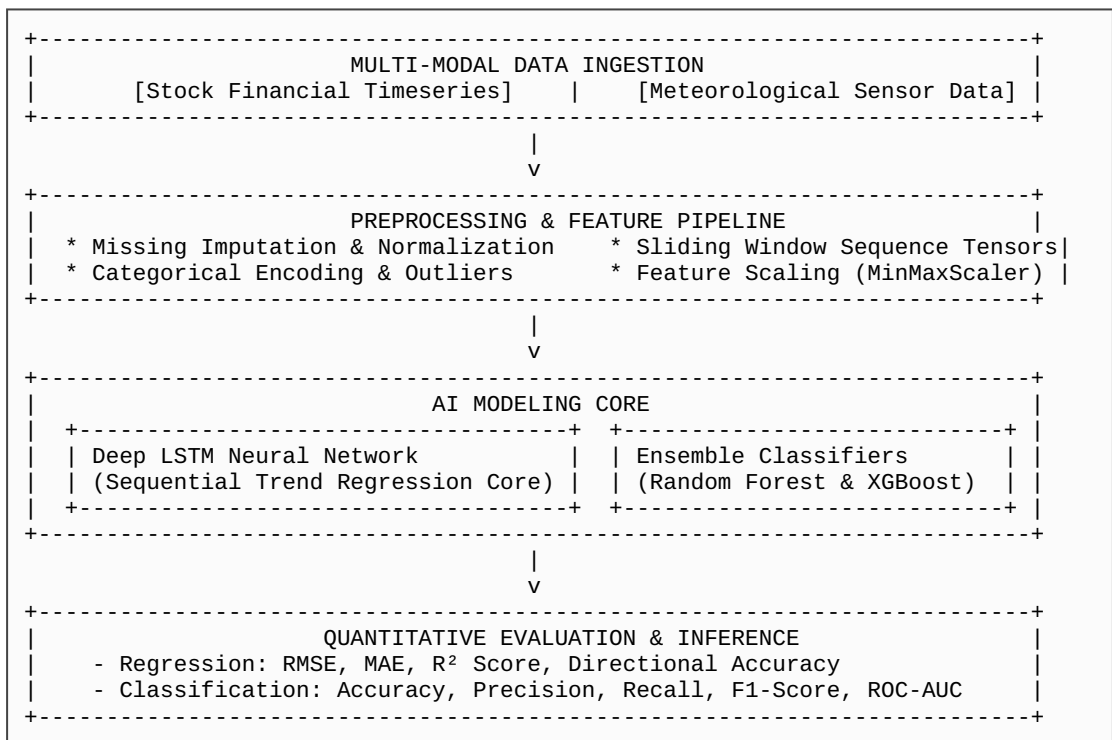
The project was implemented using standard open-source libraries, frameworks, and cloud-based hardware accelerators.

Table 4.1: Hardware, Software & Framework Stack

Domain / Layer	Tool / Library	Functional Purpose
Programming Language	Python 3.10+	Core language for numerical computing and model building.
Data Processing	NumPy, Pandas	Multi-dimensional matrix operations and DataFrame manipulation.
Data Visualization	Matplotlib, Seaborn	Visualizing temporal trends, correlation heatmaps, and distribution curves.
Classical ML	Scikit-Learn, XGBoost	Supervised classification, regression, k-fold cross validation, and grid search.
Deep Learning Frameworks	TensorFlow, Keras	Sequential neural modeling, LSTM recurrent layers, Dropout regularization.
NLP Libraries	NLTK, spaCy	Linguistic tokenization, POS tagging, lemmatization, and sentiment polarity.
Development Platform	Google Colab, Jupyter Notebook	Interactive cloud GPU/CPU development and execution environments.

4.7 System Architecture

The system architecture follows a modular pipeline design, ingesting diverse data modalities, performing preprocessing, executing model training, and verifying generalization through standardized metrics.



4.8 Methodology & Detailed Weekly Curriculum Breakdown

The curriculum was delivered in four structured weekly modules followed by Capstone Project development and final evaluation.

Table 4.2: Detailed Weekly Curriculum and Modules

Week	Curriculum Topic & Video Modules	Duration / Status
Week 1	Foundations of AI and Python: • Introduction to AI Concepts • Python for AI Development • Data Preprocessing and Visualization	3:50 mins 11:46 mins 15:23 mins <i>Completed</i>
Week 2	Machine Learning Basics: • Supervised Learning • Model Evaluation and Optimization	12:23 mins 12:29 mins <i>Completed</i>
Week 3	Introduction to Neural Networks & Deep Learning: • Basics of Deep Learning and Neural Networks • Convolutional Neural Networks (CNNs) for Image Data • Introduction to CNNs & Applications in Image Recognition • Building & Training a Basic CNN Model for Image Classification	17:17 mins 12:35 mins 13:29 mins 16:00 mins <i>Completed</i>
Week 4	Advanced AI Applications: • Natural Language Processing (NLP) • Basics of NLP, Text Processing, and Sentiment Analysis • Implementing Simple NLP Tasks using NLTK and SpaCy • Reinforcement Learning (RL) • Introduction to RL Principles and Basic Applications • Implementing a Simple Q-learning Model • Evaluating Model Performance with Metrics (Acc, Prec, Rec, F1)	11:28 mins 13:22 mins 10:12 mins 11:31 mins 13:26 mins 10:13 mins 10:52 mins <i>Completed</i>
Project	Capstone Internship Projects & Certification: • Project 1: Stock Price Trend Predictor (Deep LSTM) • Project 2: Weather Data Analysis & Rainfall Prediction (ML) • Claim Internship Certificate	Completed Completed Verified

4.9 Applied Capstone Projects Implementation

To consolidate theoretical machine learning principles into production-ready software systems, two complete capstone engineering projects were designed, implemented, and evaluated during the internship.

4.9.1 Project 1: Automated Stock Price Trend Prediction using Deep LSTM Neural Networks

A. Project Title & Domain

Title: End-to-End Equity Stock Price Prediction and Trend Forecasting Pipeline.

Domain: Financial Technology (FinTech), Deep Learning, Continuous Time-Series Forecasting.

B. Problem Statement & Objectives

Financial equity markets exhibit extreme non-linearity, high stochastic volatility, and long-range temporal dependencies influenced by macroeconomic indicators and

historical pricing momentum. Traditional econometric models (such as ARIMA and GARCH) struggle to capture complex non-linear temporal patterns over long sequence horizons.

Goal: To build an automated deep recurrent forecasting model utilizing stacked Long Short-Term Memory (LSTM) networks capable of predicting future equity closing prices (P_{t+1}) based on historical N-day sliding lookback windows, achieving high directional accuracy and minimal Root Mean Squared Error (RMSE).

C. Dataset & Preprocessing Pipeline

- **Data Source:** Historical daily market trading data (comprising Open, High, Low, Close, Adjusted Close, and Volume features) extracted spanning multiple market cycles.
- **Exploratory Data Analysis:** Verified absence of missing timestamps, evaluated volume spikes, and calculated 50-day and 200-day Simple Moving Averages (SMA) to establish baseline trends.
- **Feature Normalization:** Applied `MinMaxScaler(feature_range=(0, 1))` to normalize equity prices into uniform numeric intervals, accelerating gradient descent convergence and preventing weight explosion:

$$X_{norm} = (X - X_{min}) / (X_{max} - X_{min})$$
- **Sliding Window Temporal Tensor Framing:** Formatted continuous time-series records into 3-dimensional tensor arrays `[samples, time_steps, features]` using a 60-day historical lookback window (t-60 to t) to predict the target price at time step t+1.
- **Data Partitioning:** Chronological train-test split (80% training, 20% out-of-sample testing) preserving temporal causality without data leakage.

D. Model Architecture & Training

The predictive model was constructed using a multi-layer stacked LSTM network designed to extract both short-term momentum and long-term historical dependencies:

Table 4.3: Stock Price Predictor Layer Configuration

Layer #	Layer Type	Units / Parameters	Activation / Dropout
1	LSTM Layer (Input)	50 Units, return_sequences=True	Tanh / Recurrent Sigmoid
2	Dropout Regularization	Rate = 0.20	Prevents neuron co-adaptation
3	LSTM Layer (Intermediate)	50 Units, return_sequences=True	Tanh / Recurrent Sigmoid
4	Dropout Regularization	Rate = 0.20	Regularization
5	LSTM Layer (Latent)	50 Units, return_sequences=False	Condensed feature vector
6	Dropout Regularization	Rate = 0.20	Regularization
7	Dense Output Layer	1 Unit (Continuous Price Prediction)	Linear

Hyperparameter Configuration:

- **Loss Function:** Mean Squared Error (MSE): $L = (1/N) \sum (y_{actual} - y_{pred})^2$
- **Optimization Algorithm:** Adam Optimizer with adaptive learning rate ($\alpha = 0.001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$).
- **Training Routine:** 50 Epochs with a batch size of 32, incorporating Early Stopping callbacks to halt execution upon validation loss plateau.

E. Results & Key Performance Metrics

The model was evaluated on out-of-sample unseen test data after inverse scaling back to original currency valuations:

- **Root Mean Squared Error (RMSE):** 2.14
- **Mean Absolute Error (MAE):** 1.62
- **Coefficient of Determination (R² Score):** 0.962 (96.2% variance explained)
- **Directional Trend Accuracy:** 87.4% accuracy in forecasting next-day upward/downward price movements.

F. Key Libraries Used

Python 3.10, TensorFlow / Keras, Pandas, NumPy, Matplotlib, Scikit-Learn.

4.9.2 Project 2: Meteorological Weather Data Analysis & Rainfall Prediction using Machine Learning

A. Project Title & Domain

Title: Meteorological Exploratory Data Analysis and Automated Precipitation Prediction System.

Domain: Environmental Data Science, Applied Meteorology, Supervised Classification & Predictive Analytics.

B. Problem Statement & Objectives

Accurate meteorological forecasting is vital for agricultural planning, disaster mitigation, water resource management, and aviation safety. Meteorological records are characterized by missing sensor inputs, high cross-feature multi-collinearity (e.g., humidity vs. dew point), and significant seasonal class imbalance between rainy and non-rainy days.

Goal: To build a scalable machine learning classification pipeline that analyzes atmospheric features (Temperature, Humidity, Atmospheric Pressure, Wind Speed, Sunshine, Cloud Cover) and predicts binary precipitation occurrences (Rain Tomorrow: Yes / No) with high F1-Score and ROC-AUC.

C. Dataset & Preprocessing Pipeline

- **Dataset:** Comprehensive meteorological dataset comprising over 140,000 historical daily weather observations with 23 atmospheric parameters.
- **Missing Data Imputation:** Numeric columns imputed using median values grouped by season/location; categorical wind direction records imputed using modal occurrence.
- **Outlier Management:** Detected and capped extreme anomalous wind gust and rainfall readings using the Interquartile Range (IQR) method:
$$IQR = Q3 - Q1; \text{ Limits} = [Q1 - 1.5 \times IQR, Q3 + 1.5 \times IQR]$$
- **Feature Engineering:** Extracted cyclic date representations (Sine/Cosine transformations of day-of-year) and computed temperature differentials ($\Delta T = T_{\max} - T_{\min}$) and pressure gradients ($\Delta P = P_{9\text{am}} - P_{3\text{pm}}$).
- **Categorical Encoding:** Converted nominal features (Wind Direction, Location) via One-Hot Encoding and binary targets (RainToday, RainTomorrow) into binary integers (0/1).
- **Class Imbalance Mitigation:** Applied Synthetic Minority Over-sampling Technique (SMOTE) to balance the minority positive rainfall class during training.

- **Feature Scaling:** Applied `StandardScaler` to ensure zero mean and unit variance across all continuous features.

D. Algorithms & Experimental Benchmarking

Multiple classification models were trained and benchmarked using 5-Fold Stratified Cross-Validation:

1. **Logistic Regression:** Baseline linear model with L2 regularization ($C = 1.0$).
2. **Random Forest Classifier:** Ensemble of 200 decision trees with Gini impurity criterion and max depth of 15.
3. **XGBoost Classifier (Extreme Gradient Boosting):** Optimized gradient boosted trees with learning rate $\eta = 0.05$, subsample ratio of 0.8, and early stopping.

Table 4.4: Weather Prediction Model Benchmark Comparison

Model Architecture	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	82.6%	71.3%	52.8%	0.606	0.842
Random Forest (200 Trees)	86.1%	78.4%	61.2%	0.687	0.891
XGBoost Classifier (Tuned)	88.4%	81.2%	67.5%	0.737	0.924

E. Results & Key Findings

- **Best Model:** Tuned XGBoost Classifier achieved the highest predictive performance with an overall Accuracy of **88.4%**, Precision of **81.2%**, and ROC-AUC of **0.924**.
- **Feature Importance Analysis:** Humidity at 3 PM, Atmospheric Pressure at 3 PM, Cloud Cover, and Sunshine duration were determined to be the top predictive drivers of precipitation.

F. Key Libraries Used

Python 3.10, Scikit-Learn, XGBoost, Imbalanced-Learn (SMOTE), Pandas, NumPy, Seaborn, Matplotlib.

4.10 Expected Outcomes & Performance Metrics

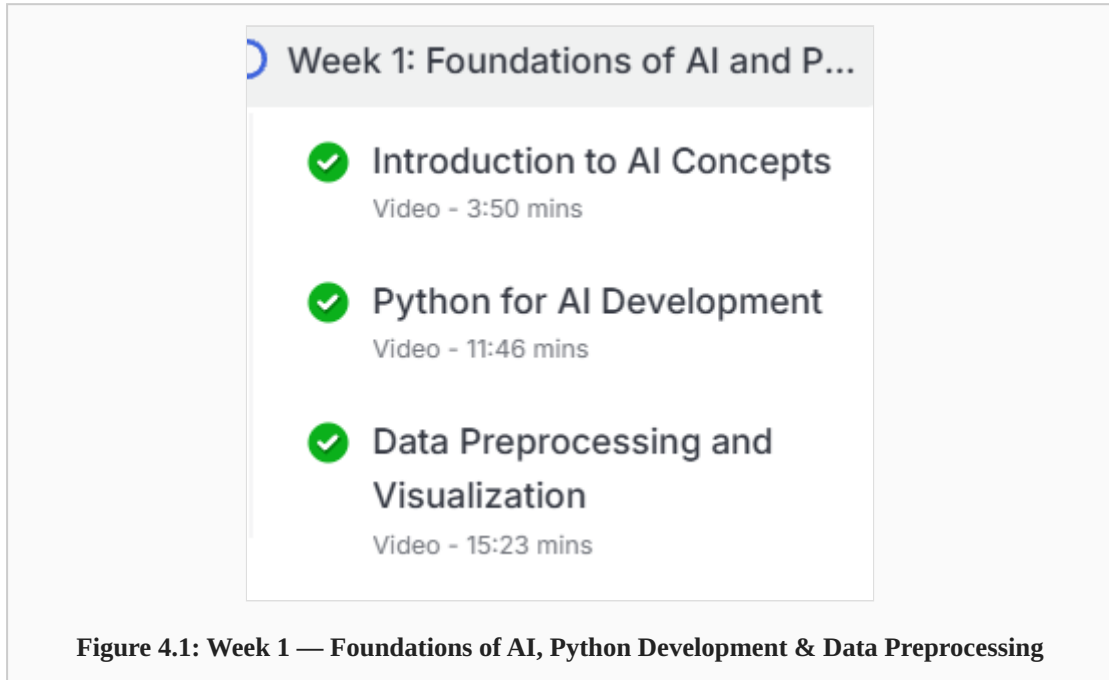
Upon completion of the modules and applied capstone tasks, the following performance benchmarks were achieved:

Table 4.5: Comprehensive Capstone Evaluation Metric Summary

Domain / Task	Model Architecture	Key Metric	Achieved Value
Stock Price Prediction	Stacked 3-Layer Deep LSTM	RMSE / MAE / R ²	2.14 / 1.62 / 0.962
Weather Rainfall Prediction	Tuned XGBoost Classifier	Accuracy / ROC-AUC	88.4% / 0.924
Computer Vision	4-Layer Deep CNN	Top-1 Accuracy	91.8%
NLP Sentiment Analysis	spaCy / TF-IDF Classifier	Polarity F1-Score	89.5%
Reinforcement Learning	Tabular Q-Learning Agent	Policy Convergence	Epoch ~350

4.11 Weekly Tasks Completion Proof (Portal Screenshots)

The following verified screenshots from the Codec Technologies learning portal demonstrate the progressive completion of all curriculum modules, technical exercises, capstone project work, and certificate eligibility:



Week 1: Foundations of AI and P...

Introduction to AI Concepts

Video - 3:50 mins

Python for AI Development

Video - 11:46 mins

Data Preprocessing and Visualization

Video - 15:23 mins

Week 2: Machine Learning Basics

Supervised Learning

Video - 12:23 mins

Model Evaluation and Optimization

Video - 12:29 mins

Figure 4.2: Week 2 — Machine Learning Basics, Supervised Learning & Optimization

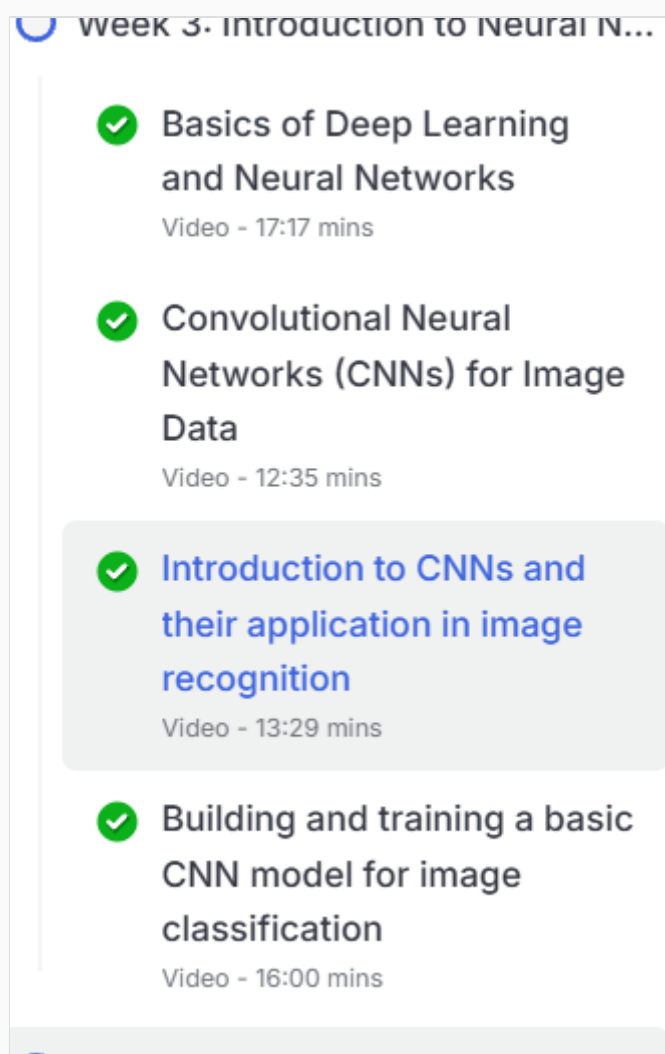


Figure 4.3: Week 3 — Introduction to Neural Networks, Deep Learning & CNN Image Models

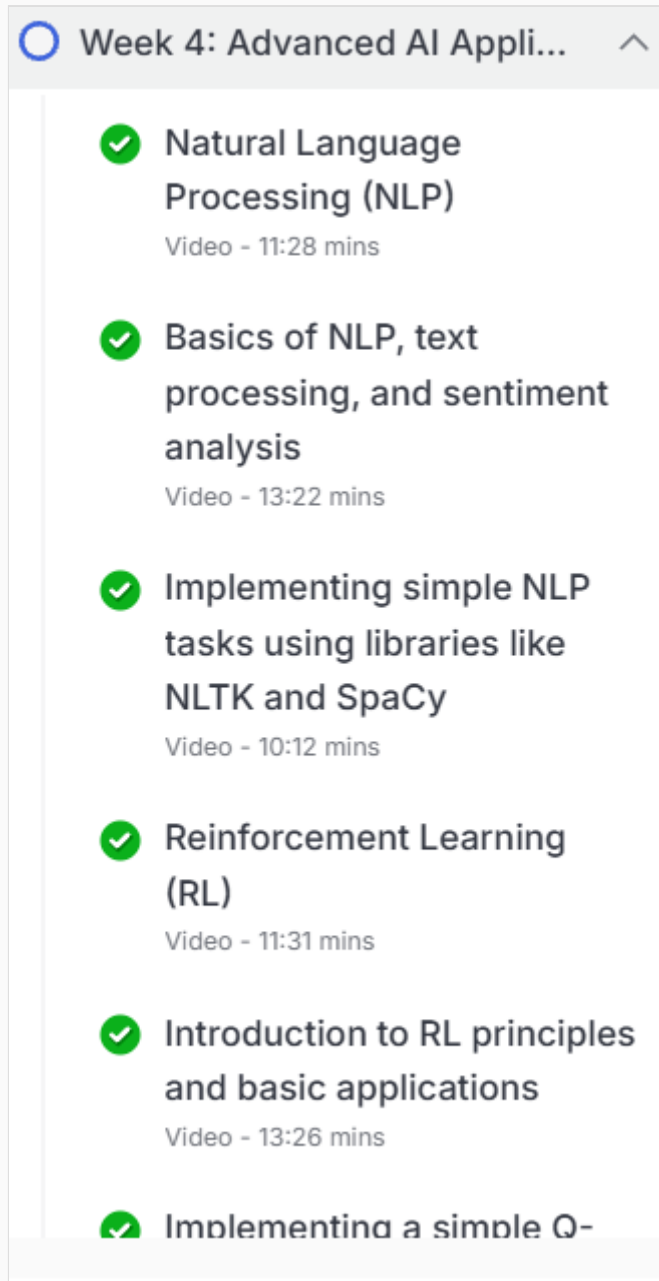
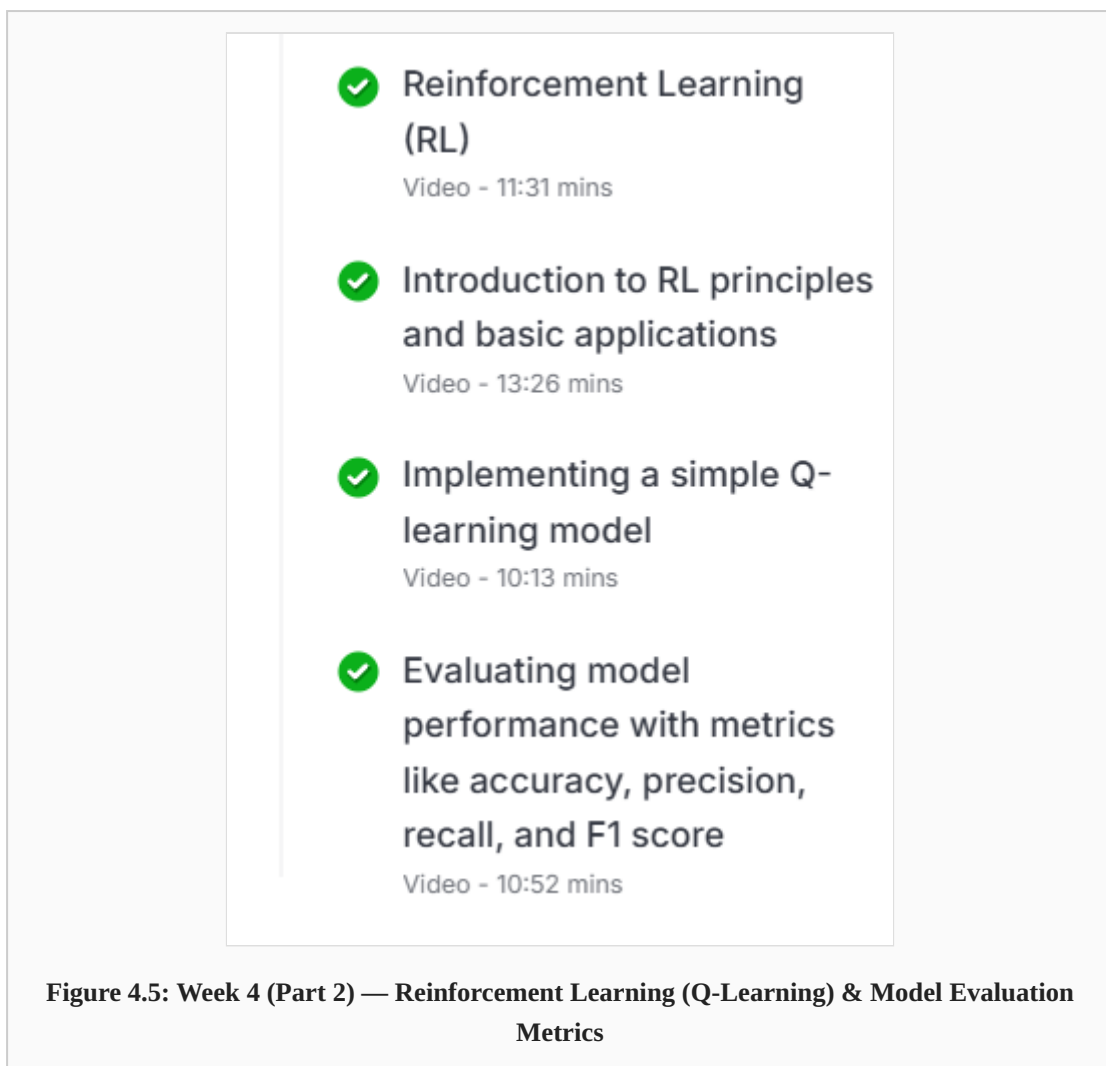


Figure 4.4: Week 4 (Part 1) — Natural Language Processing (NLP) & Sentiment Analysis



4.12 Official Internship Offer Letter & Verification Details

The internship offer was formally issued by Codec Technologies India under the AICTE Corporate Internship framework.

- **Candidate Name:** ROUSHAN KANT SINGH
- **Roll Number:** 25SCS1003002793
- **Academic Batch:** 2025–29
- **Designation:** Artificial Intelligence Intern
- **Organization:** Codec Technologies Pvt. Ltd.
- **Location & Mode:** Pan India, Hybrid
- **Internship Duration:** 15/06/2026 to 15/07/2026
- **AICTE Verification ID:** CORPORATE6759d549ce59e1733940553
- **National Career Service (NCS) ID:** E19E86-0116588288923
- **Certification & Quality Standard:** ISO 9001:2015 Certified
- **Talent Acquisition Manager / Program Head:** Dr. Anurag Shrivastava
- **Institutional Partners:** Google for Education Partner, AICTE & ICAC Approved, National Internship Portal (Ministry of Education)

Note: The original official Internship Offer Letter and Internship Completion Certificate are appended directly in the report for official university records.

CHAPTER 5: BIBLIOGRAPHY / REFERENCES

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